

Suppl 2. Calcium Data

		Pre-flight		In-flight		R+0		R+1		R+2-7		>R+7	
#	Author(s), year (units)												
	(n) Condition	Mean	$\pm \sigma$	Mean	$\pm \sigma$	Mean	$\pm \sigma$	Mean	$\pm \sigma$	Mean	$\pm \sigma$	Mean	$\pm \sigma$
		<i>d</i>	Trend	<i>d</i>	Trend	<i>d</i>	Trend	<i>d</i>	Trend	<i>d</i>	Trend	<i>d</i>	Trend

Trend legend:

(↓) authors found significant decrease from pre-flight ($p < 0.05$)

(▼) authors found trend to significant decrease from pre-flight ($0.05 < p \leq 0.1$)

(↑) authors found significant increase from pre-flight ($p < 0.05$)

(▲) authors found trend to significant increase from pre-flight ($0.05 < p < 0.1$)

(-) authors found nonsignificant change from pre-flight

(U) p-values unreported by authors.

Table S1. Data table with mean serum calcium data, standard deviation, sample size, experimental condition, and significance of results extracted from all studies.

		Pre-Flight		In-Flight		R+0		R+1		R+2-7		>R+7	
1	Smith et al. 2005 (mmol/L)												
	(6) Real	2.4	0.17	2.32	0.20	2.41	0.12	2.41	0.12	2.35	0.17	2.36	0.12
				-0.43	--	0.07	--	0.07	--	-0.29	--	-0.27	--
	(16) Real	2.41	0.24			2.44	0.36	2.44	0.04	2.37	0.44	2.36	0.24
						0.10	--	0.17	--	0.11	--	-0.21	--
2	Shackelford et al. 2004 (mg/dL)												
	(18) Simulated	9.6	1.06	9.7	0.89								
				0.10	--								
3	Linossier et al. ¹ 2022 (mg/L)	96.0	4.74	99.6	6.81					96.4	7.11		

¹ Results are reported as medians with IQRs. Standard deviations are approximated from IQRs using the formula $SD = IQR/1.35$ as given by the Cochrane Handbook for Systematic Reviews.

	(25) Simulated	2.58	0.06	2.62	0.01			2.47	0.03	2.48	0.02	2.51	0.04
				0.93	--			-2.32	↓	-2.24	↓	-1.37	--

Table S2. Data table with mean urinary calcium data, standard deviation, sample size, experimental condition, and significance of results extracted from all studies.

		Pre-Flight		In-Flight		R+0		R+1		R+2-7		>R+7	
1	Smith et al. 2005 (mmol/day)												
	(6) Real	5.3	1.4	7.9	3.8	2.7	1.7	6.5	1.4	4.7	1.7	5.3	1.0
				0.91	▲	-1.67	--	0.86	--	-0.39	--	0.00	--
	(16) Real	4.8	1.7			2.7	1.7	5.4	1.6	5.4	1.9	4.2	1.5
						-1.24	↓	0.36	--	0.36	--	-0.37	--
2	Smith et al. 2015 (mmol/day)												
	(7) Real	4.5	2.5	6.8	3.9								
				0.70	↑								
3	Zerwekh et al. 2007 (mg/kg/day)												
	(10) Simulated	116	65	180	87								
				0.83	↑								
4	Smith et al. 2003 (mmol/day)												
	(8) Simulated	5.67	3.19	7.48	3.35			5.66	2.76				
				0.55	↑			0.00	--				
5	Zwart et al. 2007 (mmol/day)												
	(7) Simulated	3.95	1.32	6.64	2.54			5.22	2.83				
				1.33	↑			0.58	--				
6	Morgan et al. 2012 (mmol/day)												
	(12) Simulated	5.5	2.1	7.0	2.5			6.3	2.0	5.7	1.8		
				0.65	↑			0.39	↑	0.10	--		

Table S3. Data table with mean ionized calcium data, standard deviation, sample size, experimental condition, and significance of results from all studies.

		Pre-Flight		In-Flight		R+0		R+1		R+2-7		>R+7	
1	Shackelford et al. 2004 (mmol/L)												
	(18) Simulated	1.2	0.01	1.3	0.01								
				1.0	↑								
2	Smith et al. 2012 (mmol/L)												
	(6) Real	1.24	0.04			1.17	0.06					1.21	0.04
						-1.37	↓					-0.75	--
3	Morgan et al. 2012 (mmol/L)												
	(12) Simulated	1.23	0.04	1.23	0.04					1.23	0.03		
				0.00	--					0.00	--		